

Organizational AI Workforce Coverage and Maturity: A Framework for Responsible AI Adoption and Governance

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Organizations are increasingly adopting artificial intelligence across diverse roles and workflows, yet existing AI maturity and governance models remain largely technology-centric and insufficiently attentive to how AI capabilities, risks, and responsibilities are distributed across the workforce. This study advances a workforce-coverage theory of organizational AI maturity that conceptualizes readiness as a function of coordinated development across three distinct domains: AI for All, representing baseline workforce enablement and safe everyday use; AI for Some, representing applied AI workflows and power users; and AI for Few, representing specialist governance, platform, and risk operations. The framework theorizes that AI risk and impact escalate across these domains and that governance intensity must scale proportionally to enable responsible and sustainable adoption. To operationalize the framework, the study introduces the AI Workforce Coverage and Maturity Survey, a structured instrument comprising 90 items organized across six subdomains per domain and measured on a standardized 0–4 maturity scale. The empirical design supports hypothesis testing through structural equation modeling, necessary-condition analysis using fuzzy-set qualitative comparative analysis, and role-based multilevel modeling. By integrating workforce segmentation, governance maturity, and balance across domains, the study contributes a generalizable theory of distributed capability maturity applicable beyond artificial intelligence to other probabilistic and analytically intensive technologies. The framework offers both theoretical advancement for information systems research and practical guidance for organizations seeking to scale complex digital technologies while maintaining accountability, equity, and control.

Keywords— Artificial Intelligence, AI Governance, AI Maturity, Responsible AI

Introduction

Despite rapid advances in artificial intelligence technologies, existing organizational AI maturity and governance frameworks remain disproportionately focused on technical capability, system performance, and tool adoption, often treating the workforce as a largely homogeneous recipient of AI change (Hansen et al., 2024; Bubicz & Ferasso, 2024; Reuel et al., 2024). While prior

research has contributed important insights into responsible AI principles, governance structures, and ethical safeguards, these models frequently under-specify how AI capabilities are unevenly distributed, exercised, and governed across different segments of the workforce (Lin, 2024; Passlack et al., 2025). As a result, organizations may appear “mature” according to aggregate metrics while simultaneously exhibiting critical misalignments between baseline AI literacy, applied AI workflows, and specialist governance capacity, leading to elevated risk, inconsistent outcomes, and stalled scaling (“Proceedings of the 20th Conference on Computer Science and Intelligence Systems (FedCSIS),” 2025; Meimandi et al., 2025). This study addresses this gap by advancing a workforce-coverage perspective on AI maturity that explicitly differentiates between AI for All, AI for Some, and AI for Few, reframing AI readiness as a socio-technical equilibrium rather than a linear progression of technical sophistication. By integrating workforce segmentation with governance maturity, the proposed framework offers explanatory power beyond existing models that emphasize either AI literacy, applied adoption, or governance in isolation, thereby responding directly to calls in the literature for more granular, human-centered, and operationally actionable approaches to responsible AI integration (Hansen et al., 2024; Reuel et al., 2024).

This study introduces a conceptual and measurement framework that categorizes organizational AI workforce engagement into three distinct domains: AI for All, AI for Some, and AI for Few. These domains represent baseline workforce enablement, applied AI use embedded in operational workflows, and specialist governance and oversight functions, respectively. Explicit differentiation of workforce coverage addresses persistent challenges in AI adoption, including fragmented implementation, uneven risk controls, and misalignment between AI capability and workforce readiness (Reuel et al., 2024; Butler et al., 2023). Many organizations demonstrate localized AI experimentation without shared expectations, governance coherence, or enterprise-level accountability. Such patterns often result in uneven value realization and elevated ethical risk, while simultaneously obscuring where targeted intervention is required. This framework elevates workforce segmentation from an implicit assumption to a primary analytic construct that can be systematically measured and managed.

The framework provides a structured approach for analyzing organizational AI maturity through the combined lenses of workforce coverage and governance alignment. It enables assessment not only of whether AI capabilities exist, but also of whether those capabilities are coherently distributed, institutionally supported, and operationally governed across the enterprise. By foregrounding balance across workforce domains, the framework reframes AI maturity as an equilibrium condition rather than a linear progression. This perspective challenges dominant stage-based maturity narratives and emphasizes relational dependencies between enablement, application, and oversight. These dependencies shape the pace, sustainability, and legitimacy of AI scaling. The study therefore advances a socio-technical model that integrates workforce capability,

applied operations, and governance capacity into a unified, empirically testable structure.

Beyond artificial intelligence, the proposed framework contributes more broadly to digital transformation theory, organizational capability maturity, and technology-enabled organizational design. By conceptualizing readiness as a function of differentiated workforce coverage rather than aggregate technological sophistication, the framework advances a generalizable model for understanding how complex, probabilistic, and distributed technologies are absorbed into organizational systems. This perspective aligns with longstanding concerns in information systems research regarding the misalignment between technical capability, human capacity, and governance structures during digital transformation initiatives. The workforce coverage lens introduced in this study is therefore not limited to AI, but is applicable to other technologies characterized by opacity, autonomy, and uneven usage patterns, including advanced analytics platforms, algorithmic decision-support systems, and large-scale data infrastructures. By foregrounding balance, interaction, and role differentiation as core explanatory mechanisms, the framework reframes capability maturity as an organizational design problem rather than a purely technological one, offering a theoretically grounded contribution that extends beyond AI governance to the broader study of socio-technical transformation in contemporary organizations.

Literature Review

Research on digital transformation consistently demonstrates that technology-centric initiatives fail when organizational readiness, leadership commitment, and human factors are underdeveloped (Anthuvan & Maheshwari, 2025; Ogot, 2025). Artificial intelligence amplifies these risks due to probabilistic outputs, opacity, and partial autonomy, which complicate accountability and oversight (Winby & Xu, 2025). Unlike deterministic systems, AI introduces uncertainty that must be actively managed through organizational processes, policies, and governance routines. As AI systems increasingly shape decisions and workflows, workforce readiness becomes a core determinant of both value realization and organizational risk exposure. Failure to account for this readiness often results in resistance, misuse, or disengagement that manifests as governance breakdowns rather than technical failure.

Existing AI maturity and adoption models have advanced understanding of technical capability development and organizational learning (Bubicz & Ferasso, 2024; Graibeh et al., 2024). However, many models emphasize system maturity while aggregating workforce considerations into generalized adoption indicators. This aggregation obscures meaningful differences in how employees interact with AI, ranging from everyday generative tool use to high-stakes governance and platform oversight (Lin, 2024; Passlack et al., 2025). These interactions carry distinct risk profiles, accountability expectations, and skill requirements. Without explicitly modeling such distinctions, maturity assessments struggle to explain why some AI initiatives scale successfully while others stagnate or fail, limiting both theoretical

explanatory power and practical utility.

Recent scholarship emphasizes AI literacy, applied capability development, and enterprise governance as distinct but interdependent dimensions of responsible AI adoption (Tjondronegoro, 2025; Hackl et al., 2025; Tabassi, 2023). Ethical AI research further underscores the need to embed responsibility and risk management within organizational processes rather than treating them as compliance add-ons (Madanchian & Taherdoost, 2025; Gandhi et al., 2025). Together, these insights point toward the necessity of integrated models that connect workforce capability with governance structures. A workforce-coverage framework responds directly to this gap by aligning differentiated roles with proportional governance and risk exposure, enabling innovation without undermining accountability.

Digital Transformation and AI Governance

Artificial intelligence represents an advanced phase of digital transformation characterized by autonomous decision logic, continuous learning, and ethical complexity (Graibeh et al., 2024; Engin, 2025). These properties challenge traditional information technology governance models that assume predictable system behavior and centralized control. AI systems evolve through data, usage, and feedback loops, often outside formal development lifecycles. As a result, governance mechanisms must adapt to dynamic risk profiles, decentralized decision-making, and emergent system behavior.

Effective AI governance encompasses ethical principles, risk management, accountability mechanisms, and strategic alignment with organizational objectives (Altepost et al., 2024; Alzahrani, 2025). Governance requirements vary significantly by workforce coverage, ranging from safe-use norms and approved tools for everyday AI use to formal evaluation, monitoring, and lifecycle controls for enterprise platforms (Schmitt, 2024; Hussein et al., 2024). Uniform governance approaches are therefore insufficient. Proportional governance aligned to workforce coverage and risk exposure is essential for sustaining innovation while maintaining organizational control and legitimacy.

Responsible AI and Risk Management

AI adoption introduces ethical, legal, and operational risks, including algorithmic bias, privacy violations, security vulnerabilities, and lack of explainability (Dubey, 2024; Gohil, 2025). These risks are not evenly distributed across an organization but are shaped by how and where AI is deployed. Responsible AI frameworks integrate ethical theory with governance controls to clarify decision responsibility and manage trade-offs under uncertainty (Madanchian & Taherdoost, 2025). Such frameworks provide normative grounding for operational decision-making and consistency across decentralized environments.

Risk exposure differs substantially across workforce segments. Baseline AI use presents diffuse, lower-impact risk, applied workflows introduce moderate to high operational risk, and enterprise AI platforms concentrate

systemic risk (Ee et al., 2024; Meimandi et al., 2025). Treating these risk profiles as equivalent leads to either excessive restriction or insufficient oversight. Workforce coverage therefore provides a practical lens for calibrating governance intensity and risk controls, improving both compliance effectiveness and organizational agility.

Workforce Enablement and Capability Maturity Models

Many AI capability maturity models emphasize advanced technical expertise while underestimating the cumulative impact of baseline AI literacy and applied operational competence (Devabhaktuni et al., 2025). This emphasis reflects an implicit assumption that value creation is driven primarily by specialists. In practice, however, everyday AI use by non-specialists shapes organizational outcomes at scale. Ignoring this reality results in blind spots that delay necessary investment in workforce development.

AI literacy across the workforce supports informed use, critical evaluation, and risk mitigation (Tjondronegoro, 2025; Kumari, 2025). Applied competencies enable workflow integration, while specialist capabilities ensure governance, platform oversight, and lifecycle management (Gupta, 2025). These capabilities are complementary rather than substitutable. An effective maturity model must therefore articulate differentiated capability pathways aligned with workforce coverage, enabling targeted investment, clearer accountability, and sustainable organizational learning.

Gaps in Existing Models and the Need for a Coverage-Based Framework

Most existing AI maturity models fail to adequately differentiate workforce roles and risk exposure (Gandhi et al., 2025; Reuel et al., 2024). As a result, they often conflate adoption with readiness and maturity with deployment scale. Without explicit segmentation, organizations risk overinvesting in advanced systems while neglecting baseline enablement or governance capacity (Onnen, 2025). Such imbalance increases ethical risk, operational fragility, and resistance to adoption. A coverage-based framework directly addresses these limitations by aligning workforce engagement, governance structures, and capability development.

Conceptual Framework

Table 1. Organizational AI Workforce Coverage and Maturity Framework

Domain	Workforce Scope	Primary Focus	Typical Risks	Governance Emphasis
AI for All	Entire workforce	Baseline AI literacy and safe everyday use	Diffuse misuse, data leakage, quality errors	Shared norms, policies, approved tools
AI for Some	Power users and applied teams	Workflow automation, analytics, decision support	Undocumented logic, bias propagation	Human review, documentation, testing
AI for Few	Specialists and leaders	Governance, platforms, security, lifecycle oversight	Systemic failure, regulatory exposure	Formal controls, monitoring, accountability

Table 1 provides a structured overview of the three workforce coverage domains and their associated characteristics, serving as the conceptual foundation for the propositions and hypotheses developed later in the manuscript. Each domain is theorized to map directly to distinct mechanisms of organizational capability, risk exposure, and governance demand, which are subsequently formalized through testable propositions and empirically operationalized using the survey instrument. As organizations move from baseline AI enablement to applied workflow integration and, ultimately, to enterprise-level governance and platform operations, both the potential impact and the associated risk profile of AI use increase in scope, complexity, and consequence. Accordingly, the framework posits that governance intensity must scale proportionally across domains, shifting from normative guidance and safe-use expectations in AI for All, to applied controls and human review in AI for Some, and to formalized accountability, monitoring, and lifecycle management in AI for Few. This escalation logic underpins the central claim of the framework that responsible AI adoption depends not on isolated domain maturity, but on coordinated development and balance across workforce segments.

AI for All: Baseline Workforce Enablement

The AI for All domain represents the foundational layer of organizational AI maturity and focuses on establishing a shared baseline of AI literacy, expectations, and safe-use norms across the entire workforce. This domain is concerned with ensuring that all employees, regardless of role or technical background, possess a sufficient understanding of what AI systems can and cannot do, how AI-generated outputs should be interpreted, and what ethical, legal, and organizational constraints govern AI use (Passlack et al., 2025; Tadimalla & Maher, 2024; Long & Magerko, 2020). The objective is not to cultivate technical expertise, but to foster informed, critical, and responsible engagement with AI technologies that are increasingly embedded in everyday

work activities. Prior research in human-centered AI and information systems emphasizes that literacy at this level is a prerequisite for meaningful human oversight and trust calibration (Rai, 2020; Siau & Wang, 2018).

Baseline workforce enablement plays a critical role in shaping how AI diffuses throughout an organization. In the absence of shared literacy and expectations, AI adoption tends to occur informally and unevenly, often driven by individual experimentation rather than institutional strategy. This pattern is well documented in socio-technical adoption research, which shows that decentralized experimentation without shared norms increases variability in outcomes and risk exposure (Davenport & Ronanki, 2018; Kellogg et al., 2020). Such conditions can lead to inconsistent practices, unmanaged data risk, and erosion of accountability, particularly when general-purpose AI tools are used for decision support, content generation, or analysis. By contrast, organizations that invest in baseline enablement are better positioned to define acceptable-use boundaries, promote verification practices, and ensure that AI outputs are treated as decision inputs rather than authoritative conclusions.

Importantly, the AI for All domain also serves as a cultural and normative foundation for responsible AI use. It encompasses not only technical awareness, but also shared values related to transparency, accountability, fairness, and data stewardship. Employees operating within this domain should understand when human judgment is required, how to recognize potential bias, hallucination, or overgeneralization in AI outputs, and how to escalate concerns or uncertainties (Eubanks, 2018; Rai, 2020). These capabilities support psychological safety and responsible experimentation, allowing organizations to encourage innovation without normalizing uncritical reliance on automated outputs. Research in algorithmic accountability highlights that many harms arise not from malicious intent, but from routine, uninformed use by well-meaning individuals (Raji et al., 2020).

From a governance perspective, baseline enablement functions as a first line of defense against diffuse organizational risk. Many AI-related incidents originate not from advanced models or complex pipelines, but from routine misuse, inappropriate data input, or misunderstanding of system limitations. Establishing baseline literacy and expectations reduces this risk surface by aligning everyday AI use with organizational standards, ethical commitments, and regulatory obligations (OECD, 2019; ISO/IEC, 2023). As such, AI for All is a necessary precondition for more advanced AI capabilities, providing the shared understanding required for coherent and responsible scaling.

As early as 2009, nation-state actors have been gaining access to U.S. power grids through methods like compromised business networks and phishing campaigns. The goal of these intrusions was not immediate disruption, but silent, strategic reconnaissance aimed at planting backdoors and mapping control systems for future use. This approach of "gain access now and weaponize later" is designed to build long-term political and military leverage, allowing an adversary to apply cyber pressure during a future

geopolitical crisis without firing a single missile. The pace of these operations increased significantly with the development of Russia's BlackEnergy malware toolkit. This same toolkit, used by the GRU-affiliated Sandworm group to disable parts of Ukraine's electrical grid, was discovered dormant inside U.S. substations in the Midwest and Pacific Northwest as early as 2014. Although the exploits were not active, their presence represented a form of strategic prepositioning, a digital equivalent to parking submarines off an adversary's coast. The persistence of this malware is enabled by fragile security practices at U.S. utilities, such as unpatched firmware, and a lack of forensic visibility that allows attackers to "live off the land" using default system tools to avoid detection.

AI for Some: Power Users and Applied Workflows

The AI for Some domain encompasses employees who move beyond general awareness and actively integrate AI into applied workflows to enhance productivity, analysis, and decision-making (Zary, 2025). These individuals, often described as power users, operate at the intersection of domain expertise and AI capability. Their work typically involves automating processes, augmenting analytical tasks, generating insights, or supporting operational decisions through AI-enabled tools. Prior research characterizes this mode of work as human-AI symbiosis, in which AI augments rather than replaces professional judgment (Jarrahi, 2018; Shrestha et al., 2019).

Applied AI use introduces a higher degree of variability and organizational impact than baseline use. Power users make consequential choices about which AI tools to deploy, how to structure inputs, how to interpret outputs, and how to embed AI-generated information into downstream decisions. These choices require applied competencies related to data readiness, configuration or prompt design, contextual interpretation, and ethical judgment (Dubey, 2024; Silva et al., 2024). Errors at this level can propagate across workflows, affecting customers, employees, or strategic outcomes. Organizational research on algorithmic work shows that even small misalignments at this layer can scale rapidly when AI is embedded into routine processes (Kellogg et al., 2020).

Because of this elevated impact, the AI for Some domain requires more structured governance and enablement than baseline literacy alone can provide. Applied AI workflows benefit from explicit use-case prioritization, documentation standards, human review thresholds, testing protocols, and rollback mechanisms. These practices align with research on responsible deployment, which emphasizes that governance must be embedded within workflows rather than imposed externally (Shrestha et al., 2019; Weber et al., 2024). At the same time, governance mechanisms must remain proportionate to avoid suppressing innovation or discouraging experimentation among skilled practitioners.

The AI for Some domain also plays a critical intermediary role within the broader framework. Power users frequently act as translators between strategic intent and operational reality, identifying both opportunities for AI-driven improvement and limitations encountered in practice.

Their experiences provide essential feedback for refining governance policies, platform capabilities, and training programs. In this way, the AI for Some domain functions not only as an execution layer, but also as an organizational learning mechanism through which AI maturity evolves iteratively (Davenport & Ronanki, 2018; Jarrahi et al., 2023).

AI for Few: Specialist Governance, Platform, and Risk Operations

The AI for Few domain comprises a relatively small group of specialists responsible for enterprise-level AI governance, platform architecture, security, compliance, and lifecycle management (Schmitt, 2024). These roles include AI governance leads, data and AI architects, security and privacy professionals, model risk managers, and senior leaders accountable for AI strategy. Although limited in number, this domain carries disproportionate responsibility due to its role in defining the rules, infrastructure, and oversight mechanisms that shape all other AI use within the organization. Organizational science research consistently shows that centralized governance capacity is essential for coordinating complex, cross-cutting technologies such as AI (Weber et al., 2024).

Specialists in this domain are tasked with establishing and maintaining the operating system for responsible AI. This includes defining governance models, risk tiering schemes, evaluation and approval processes, monitoring and incident response mechanisms, and compliance with regulatory and ethical standards (Tabassi, 2023; Gandhi et al., 2025; ISO/IEC, 2023). Their work ensures that AI systems are not only effective, but also auditable, explainable, secure, and aligned with organizational values. Research on AI accountability emphasizes that without such lifecycle controls, organizations are unable to detect drift, bias, or misuse once systems are deployed (Raji et al., 2020).

The AI for Few domain is also responsible for platform-level decisions that influence scalability and interoperability. Choices about deployment architectures, data access controls, tooling standards, and lifecycle management directly affect how easily AI capabilities can be extended across the organization. Weaknesses in this domain often manifest as fragmented tooling, duplicated effort, or inability to monitor AI behavior over time. By contrast, strong specialist capacity enables organizations to scale AI confidently while maintaining oversight, security, and compliance (OECD, 2019; NIST, 2023).

Critically, the AI for Few domain anchors organizational trust in AI systems. Employees, customers, regulators, and external stakeholders rely on the assurance that AI-enabled decisions are governed responsibly and that accountability mechanisms exist when harm occurs. When governance capacity is insufficient, trust erodes quickly, even if technical performance appears strong. For this reason, the AI for Few domain functions not merely as a control layer, but as a strategic enabler of sustainable and legitimate AI adoption across the enterprise.

Interaction of Domains and Risks of Imbalance

Imbalanced development across domains produces characteristic failure patterns, such as applied AI without governance or governance without workforce enablement. These imbalance signatures explain why organizations with similar technical capabilities experience divergent outcomes. Balanced maturity across all three domains is associated with reduced incident rates, higher trust, and sustained value creation (Wolfe et al., 2025). Identifying imbalance early enables strategic intervention rather than reactive remediation.

Boundary Conditions and Scope of Applicability

The AI Workforce Coverage and Maturity framework is designed to explain organizational readiness and governance alignment in contexts where artificial intelligence is deployed across multiple roles, workflows, and decision layers. Accordingly, several boundary conditions shape the scope of its applicability.

First, the framework assumes a minimum level of organizational scale and role differentiation. It is most applicable to mid-sized and large organizations in which AI use spans baseline employee interactions, applied workflows, and specialist governance or platform functions. In very small organizations or early-stage startups, where AI responsibilities are highly centralized or informal, the distinctions between AI for All, AI for Some, and AI for Few may be less pronounced and require adaptation.

Second, the framework is intended for environments in which AI is embedded in operational, analytical, or decision-support processes rather than limited to experimental or isolated technical deployments. Organizations using AI primarily for narrow automation or proof-of-concept initiatives may not yet exhibit the workforce coverage dynamics that the framework is designed to capture.

Third, the framework is most relevant in institutional and regulatory contexts where ethical, legal, and accountability considerations are salient. While the model is applicable across sectors, its explanatory power is strongest in domains such as education, healthcare, public administration, and regulated industries, where governance capacity and risk containment are integral to sustainable AI scaling.

Finally, the framework reflects a snapshot of organizational maturity at a given point in time. While it is compatible with longitudinal analysis, the present study does not model temporal transitions between maturity states. Future research may extend the framework to examine dynamic progression and path dependency in workforce coverage and governance alignment. All propositions and hypotheses that follow are evaluated within these defined boundary conditions.

Propositions and Hypotheses for Empirical Validation

The AI Workforce Coverage and Maturity framework advances the central claim that responsible AI adoption and scaling are fundamentally socio-technical phenomena shaped by how AI capabilities are distributed, governed, and operationalized across the workforce. Rather than conceptualizing AI maturity as a unidimensional progression of technological sophistication, the framework theorizes AI maturity as an equilibrium condition that emerges from the coordinated development of baseline workforce enablement, applied AI workflows, and specialist governance and lifecycle operations.

This section formally articulates the theoretical propositions derived from the conceptual framework and translates them into empirically testable hypotheses. Each hypothesis is explicitly mapped to survey domains, subdomains, and representative item sets from the AI Workforce Coverage and Maturity Survey. This mapping establishes construct validity, clarifies analytic intent, and demonstrates how abstract theoretical claims are operationalized through observable organizational practices.

Structural equation modeling was employed as the primary analytic technique to test the hypothesized relationships derived from the AI Workforce Coverage and Maturity framework. SEM was selected due to its ability to simultaneously estimate latent constructs, account for measurement error, and evaluate complex structural relationships in a theory-driven manner. All models were specified a priori, consistent with confirmatory SEM practices, and no post hoc model modifications were introduced.

To ensure analytic transparency and reproducibility, the structural models corresponding to each hypothesis are documented explicitly in tabular form. Tables X through X+6 specify, for each hypothesized model, the endogenous and exogenous constructs, structural paths, expected directional effects, and the exact survey item groupings used to operationalize each latent variable. These tables function as formal SEM path specifications and collectively replace the need for informal graphical diagrams, while preserving complete model clarity.

Workforce Coverage Balance and Governance Effectiveness

The framework posits that AI governance effectiveness does not arise from isolated excellence in any single workforce domain, but from balanced maturity across all three domains. Organizations that invest heavily in applied AI use without corresponding investment in baseline enablement or specialist governance often experience governance gaps, inconsistent outcomes, and heightened risk. Conversely, organizations with strong governance capacity but weak applied enablement may fail to translate policy into operational value. Balanced maturity is therefore theorized as a higher-order organizational capability.

Proposition 1.

Organizations exhibiting balanced maturity across AI for All, AI for Some, and AI for Few domains demonstrate higher overall AI governance effectiveness than organizations with uneven domain development. This proposition is operationalized through the following hypotheses:

H1a. Balanced maturity across AI for All, AI for Some, and AI for Few domains is positively associated with overall AI governance effectiveness.

H1b. Greater variance between domain maturity scores is negatively associated with perceived AI reliability, trustworthiness, and consistency.

Table 2. Survey Mapping for H1a–H1b:

Element	Mapping
Domains	AI for All, AI for Some, AI for Few
Subdomains	A1–A6, S1–S6, F1–F6
Key Items	A28–A30 (benefits, adoption), S26–S30 (workflow outcomes), F27–F30 (monitoring, improvement)
Construct	Balance operationalized as inverse variance across domain scores

Table 3. SEM Structural Model for Workforce Coverage and AI Governance Effectiveness (Aligned to H1a)

Component	Specification
Hypothesis	H1a: Balanced maturity across AI for All, AI for Some, and AI for Few is positively associated with AI governance effectiveness
Endogenous Construct	AI Governance Effectiveness (AIGE)
Exogenous Constructs	AI for All (AFA); AI for Some (AFS); AI for Few (AFF)
Structural Paths	AFA → AIGE; AFS → AIGE; AFF → AIGE
Expected Signs	$\beta_1 > 0$; $\beta_2 > 0$; $\beta_3 > 0$
AI for All Indicators	A1–A5 (culture and expectations); A6–A10 (policy and risk awareness); A11–A15 (quality and verification); A21–A25 (skills and training); A28–A30 (benefits and improvement)
AI for Some Indicators	S1–S5 (use-case identification); S6–S10 (governance and human review); S12–S20 (documentation, testing, rollback); S26–S30 (outcomes and improvement)
AI for Few Indicators	F1–F4 (operating model and accountability); F9–F15 (evaluation and compliance); F27–F30 (monitoring and lifecycle improvement)
Covariances	AFA ↔ AFS; AFA ↔ AFF; AFS ↔ AFF
Estimation Method	Maximum likelihood SEM
Purpose	Establish baseline explanatory validity of workforce-segmented AI maturity

The baseline structural model tests the direct effects of workforce coverage domains on AI governance effectiveness (Table 3). In this model, AI for All, AI for Some, and AI for Few are specified as correlated exogenous latent constructs predicting a latent governance effectiveness outcome. This model establishes the foundational explanatory validity of the workforce-segmented maturity framework prior to introducing balance or interaction effects.

Table 4. Second-Order SEM for Workforce Coverage Balance and Outcome Stability (Aligned to H1b and H4)

Component	Specification
Hypotheses	H1b: Variance between domains negatively associated with governance outcomes; H4: Imbalance associated with outcome variability
Second-Order Construct	Workforce Coverage Balance
First-Order Constructs	AI for All; AI for Some; AI for Few
Outcome Construct	AI Outcome Stability (AIOS)
Structural Paths	AFA → Balance; AFS → Balance; AFF → Balance; Balance → AIOS
Alternative Imbalance Specification	AIOS ←
AI Outcome Indicators	S26–S30 (applied workflow outcomes); F27–F30 (monitoring, incident learning)
Expected Signs	$\lambda_1\text{--}\lambda_3 > 0$; $\beta_1 > 0$; imbalance coefficient > 0
Estimation Method	Second-order SEM; robustness check via imbalance score
Purpose	Model balance explicitly as a higher-order maturity construct

Workforce coverage balance is modeled using two complementary approaches. First, a second-order latent construct captures shared variance across AI for All, AI for Some, and AI for Few, representing coordinated maturity development. Second, imbalance is operationalized as the absolute difference between applied AI maturity and governance maturity. This dual approach follows organizational design literature emphasizing both alignment and misalignment effects and allows sensitivity analysis across linear and configurational assumptions.

Balance across workforce coverage domains is modeled explicitly in a second-order SEM (Table 4, H1b and H4). In this specification, AI for All, AI for Some, and AI for Few load onto a higher-order latent construct representing workforce coverage balance, which in turn predicts AI outcome stability. As a robustness check, imbalance is also operationalized using an absolute difference score between applied AI maturity and governance maturity. This dual specification allows assessment of both coordinated maturity and misalignment effects, consistent with organizational design and systems theory.

Baseline Enablement as a Necessary Condition for Applied AI Scaling

The framework asserts that baseline workforce enablement provides the cognitive, ethical, and procedural foundation upon which applied AI workflows depend. Without shared expectations, safe-use norms, and basic AI literacy, applied AI efforts are unlikely to scale coherently or responsibly. Importantly, the framework does not assert that baseline enablement alone is sufficient, but rather that it is a necessary precondition.

Proposition 2.

Baseline workforce enablement is a necessary but insufficient condition for the effective scaling of applied AI workflows. This proposition yields the following hypotheses:

H2a. AI for All maturity is positively associated with AI for Some maturity.

H2b. Organizations with low AI for All maturity do not exhibit high AI for Some maturity, regardless of investment in applied AI tools.

Table 5. Survey Mapping for H2a–H2b

Element	Mapping
Domains	AI for All → AI for Some
Subdomains	A1 (culture), A2 (policy), A5 (training); S1 (use cases), S5 (enablement)
Key Items	A1–A5, A21–A25; S1–S5
Analytic Logic	Necessary-condition testing using fsQCA

Table 6. SEM Enablement Model for Baseline and Applied AI Maturity

Component	Specification
Hypothesis	H2a: AI for All maturity is positively associated with AI for Some maturity
Endogenous Construct	AI for Some
Exogenous Construct	AI for All
Structural Path	AFA → AFS
Expected Sign	$\beta_1 > 0$
AI for All Indicators	A1–A5; A6–A10; A21–A25
AI for Some Indicators	S1–S5; S6–S10; S12–S20
Estimation Method	SEM (association testing)
Complementary Method	fsQCA for necessary-condition testing (H2b)
Purpose	Establish baseline enablement relationship prior to necessity analysis

The enablement relationship between baseline workforce maturity and applied AI workflows is tested using a parsimonious SEM specification (Table 6, H2a). AI for All

is modeled as an exogenous predictor of AI for Some, establishing the directional association necessary for subsequent necessary-condition testing. While SEM is used to estimate association strength, the necessity logic underlying H2b is evaluated using fuzzy-set qualitative comparative analysis, as described in the analytic strategy.

Governance Capacity and Risk Containment

As AI systems are embedded into operational workflows, organizational exposure to ethical, legal, and operational risk increases. The framework theorizes that specialist governance capacity functions as a risk dampener, moderating the relationship between applied AI use and adverse outcomes. Governance maturity is therefore positioned not merely as oversight, but as an enabling control structure.

Proposition 3.
Specialist governance maturity mitigates the operational and ethical risks associated with applied AI workflows. This proposition is operationalized through moderation hypotheses:

H3a. AI for Few maturity is negatively associated with the frequency of AI-related operational and ethical incidents.

H3b. AI for Few maturity moderates the relationship between AI for Some maturity and AI-related risk, such that higher governance maturity weakens the positive relationship between applied AI use and risk exposure.

Table 7. Survey Mapping for H3a–H3b

Element	Mapping
Domains	AI for Some × AI for Few
Subdomains	S2 (governance), S3 (data readiness); F2 (assurance), F3 (security), F6 (monitoring)
Key Items	S6–S20; F9–F15, F27–F30
Analytic Method	SEM moderation analysis

Table 8. Moderated SEM for Governance as Risk Dampener (Aligned to H3a and H3b)

Component	Specification
Hypotheses	H3a: Governance maturity reduces AI risk; H3b: Governance moderates applied AI risk
Endogenous Construct	AI Risk Exposure
Exogenous Constructs	AI for Some; AI for Few
Interaction Term	AI for Some × AI for Few
Structural Paths	AFS → AIR; AFF → AIR; AFS×AFF → AIR
AI Risk Indicators	S6–S10 (human review); S16–S20 (testing and rollback); F9–F15 (assurance, security, compliance); F27–F30 (incident monitoring)
Expected Signs	$\beta_1 > 0$; $\beta_2 < 0$; $\beta_3 < 0$
Estimation Method	Latent interaction SEM
Purpose	Test governance capacity as a risk-containment mechanism

Interaction effects are modeled using latent variable interaction techniques to avoid attenuation from measurement error. The interaction term captures the conditional effect of applied AI maturity on risk exposure as a function of governance maturity. A negative interaction coefficient indicates that governance capacity dampens the risk associated with increased applied AI use, consistent with responsible AI governance theory.

Governance capacity as a risk-containment mechanism is evaluated through a moderated SEM specification (Table 8). In this model, AI-related risk exposure is regressed on applied AI maturity, governance maturity, and their latent interaction. Latent interaction modeling is used to avoid attenuation from measurement error and to accurately capture the conditional effect of applied AI use as a function of governance capacity.

Imbalance Signatures and Outcome Variability

Beyond average effects, the framework emphasizes imbalance signatures as structural explanations for AI failure. Specifically, organizations with high applied AI capability but low governance maturity are expected to experience greater variability in AI outcomes due to inconsistent controls, undocumented logic, and unmanaged drift.

Proposition 4.
Imbalance between applied AI capability and governance capacity is associated with increased variability in AI outcomes. This proposition is tested through the following hypothesis:

H4. Greater absolute difference between AI for Some maturity and AI for Few maturity is positively associated with variability in AI-enabled outcomes.

Table 9. Survey Mapping for H4

Element	Mapping
Domains	AI for Some vs AI for Few
Subdomains	S6 (outcomes); F6 (monitoring)
Key Items	S26–S30; F27–F30
Construct	Absolute difference score

Table 4 provides second-order SEM for workforce coverage balance and outcome stability.

Workforce Segmentation and Measurement Validity

A core methodological claim of the framework is that AI maturity cannot be validly assessed using aggregated measures that collapse heterogeneous workforce experiences. Explicit workforce segmentation is therefore theorized to improve explanatory power and interpretive validity.

Proposition 5.

AI maturity assessments that differentiate workforce coverage domains provide greater explanatory power than aggregated AI maturity models. This proposition yields the following hypothesis:

H5. A workforce-segmented AI maturity model explains more variance in organizational AI outcomes than a single aggregated AI maturity index.

Table 10. Survey Mapping for H5

Element	Mapping
Domains	All domains
Subdomains	All
Key Items	All items
Comparison	Segmented model vs aggregated index

Table 11. SEM Model Comparison for Workforce-Segmented vs Aggregated Maturity (Aligned to H5)

Component	Aggregated Model	Workforce-Segmented Model
Hypothesis	H5	H5
Predictor(s)	AI Maturity Index (mean of all items)	AI for All; AI for Some; AI for Few
Outcome	AI Governance and Outcome Measures	AI Governance and Outcome Measures
Structural Paths	Index → Outcome	AFA → Outcome; AFS → Outcome; AFF → Outcome
Indicators	All A, S, F items aggregated	Domain-specific item sets
Fit Comparison	χ^2 , CFI, RMSEA	χ^2 , CFI, RMSEA
Evaluation Criteria	Δ CFI; $\Delta\chi^2$; R^2	Δ CFI; $\Delta\chi^2$; R^2
Purpose	Demonstrate explanatory advantage of workforce segmentation	

To test the added explanatory value of workforce segmentation, a model comparison approach is employed (Table 11). An aggregated AI maturity index model is estimated and compared against a workforce-segmented model using chi-square difference testing, changes in comparative fit indices, and explained variance. This comparison directly evaluates whether differentiating workforce coverage domains improves explanatory power relative to single-index maturity models.

Role-Based Perceptual Divergence

Because AI maturity is enacted through role-specific practices, the framework anticipates systematic differences in how maturity is perceived across organizational roles. These differences are not treated as error, but as theoretically meaningful signals of governance alignment or misalignment.

Proposition 6.

Perceptions of AI maturity differ systematically across organizational roles. This proposition is operationalized as:

H6. Reported AI maturity differs significantly by organizational role, with frontline and applied users reporting higher maturity in AI for All and AI for Some domains than governance respondents.

Table 12. Survey Mapping for H6

Element	Mapping
Domains	AI for All, AI for Some
Subdomains	A1–A6; S1–S6
Key Items	Role- and perspective-segmented responses
Analytic Method	Multigroup SEM or multilevel modeling

Table 13. Multigroup SEM for Role-Based Perceptual Divergence (Aligned to H6)

Component	Specification
Hypothesis	H6: AI maturity perceptions differ by organizational role
Groups	Executives; IT/Security; Applied Users; Governance Specialists
Constructs Tested	AI for All; AI for Some; AI for Few
Indicators	Domain-specific item sets (A#, S#, F#)
Invariance Tests	Configural; Metric; Scalar
Comparison Metrics	Δ CFI; Δ RMSEA
Outcome	Differences in latent means and path strengths
Purpose	Treat perceptual divergence as theoretically meaningful signal

Role-based perceptual divergence is examined using multigroup SEM (Table 13). Identical structural models are estimated across respondent role groups, including executive leadership, IT and security professionals, applied users, and governance specialists. Measurement invariance is assessed sequentially prior to group comparison, and differences in latent means and path coefficients are interpreted as theoretically meaningful indicators of governance alignment or misalignment.

Governance Maturity and Equity-Aligned Practices

In education, public service, and regulated contexts, responsible AI adoption requires alignment with equity and accessibility commitments. The framework theorizes that governance maturity enables such alignment by embedding accountability, review, and override mechanisms.

Proposition 7.

Governance maturity is positively associated with equity- and accessibility-aligned AI practices. This proposition is tested through the following hypothesis:

H7. AI for Few maturity is positively associated with equity- and accessibility-aligned AI practices in student-facing and public service environments.

Table 14. Survey Mapping for H7

Element	Mapping
Domains	AI for Few + Equity module
Subdomains	F2–F6; E1–E12
Key Items	F11–F30; E1–E12
Outcome	Equity and accessibility alignment

Table 15. SEM for Governance Maturity and Equity-Aligned AI Practices (Aligned to H7)

Component	Specification
Hypothesis	H7: Governance maturity positively associated with equity-aligned AI practices
Endogenous Construct	Equity and Accessibility Practices
Exogenous Construct	AI for Few
Structural Path	AFF → EAP
AI for Few Indicators	F2 (evaluation); F11–F15 (compliance and procurement); F27–F30 (monitoring and improvement)
Equity Indicators	E1–E12 (accessibility review, bias review, explainability, feedback mechanisms)
Expected Sign	$\beta_1 > 0$
Estimation Method	SEM
Purpose	Link governance maturity to equity and accessibility outcomes

The relationship between governance maturity and equity- and accessibility-aligned AI practices is tested using a focused SEM specification (Table 14). In this model, AI for Few maturity predicts a latent construct capturing equity and accessibility practices derived from the optional survey module. This specification links enterprise governance capacity to downstream equity outcomes in regulated and student-facing environments.

Together, these propositions and hypotheses provide a direct and transparent bridge between the conceptual framework and empirical testing. By explicitly mapping theoretical claims to survey constructs, the study establishes a clear analytic pathway that supports rigorous hypothesis testing while preserving the socio-technical complexity of organizational AI maturity.

The SEM specifications provide a complete, hypothesis-aligned analytic framework that preserves clear separation between measurement and structure while enabling rigorous testing of balance, moderation, and segmentation effects central to the theoretical model. All structural models were fully specified prior to estimation and evaluated without post hoc modification.

To address the research question, this study adopted a mixed-methods approach that integrates large-scale quantitative data analysis with in-depth qualitative case the evolving landscape of state-sponsored cyber conflict.

Methods

This study adopts a cross-sectional, multi-respondent survey design intended to empirically test the AI Workforce Coverage and Maturity framework across heterogeneous organizational contexts. The design is explanatory and theory-testing in nature, following a prior conceptual development and instrument construction phase. The primary

objective is to examine how differentiated workforce coverage domains, their balance, and their interaction are associated with AI governance effectiveness, risk outcomes, and equity-aligned practices.

The unit of analysis is the organization, while the unit of observation is the individual respondent. This distinction is critical given that AI maturity and governance are organizational properties but are perceived and enacted through role-specific practices. To address this, the study intentionally collects responses from individuals occupying diverse functional roles, including executive leadership, IT and information security, analytics and AI specialists, operational managers, faculty or frontline staff, and governance or compliance professionals. Respondents are asked to indicate both their role and their perspective, defined as whether they are responding based on unit-level experience or enterprise-level visibility. This design supports both role-based segmentation and aggregation to organizational-level constructs.

The study is designed to support both single-respondent and multi-respondent organizational participation. Where multiple respondents from the same organization are available, responses are aggregated using theoretically justified rules, such as role-weighted means or consensus thresholds, and analyzed using multilevel techniques. Where only one respondent per organization is available, analyses are conducted at the organizational level with robustness checks for role-based bias. This flexibility increases external validity while preserving analytic rigor.

Several design features are incorporated to mitigate common threats to validity in self-reported organizational surveys. First, survey items are framed around observable, routine practices rather than aspirational goals or future plans, reducing social desirability bias. Second, anchored maturity descriptors constrain interpretation and promote consistent scale usage across respondents. Third, role-based segmentation allows identification of systematic perceptual divergence, which is treated as a theoretically meaningful phenomenon rather than noise. Finally, open-ended items are included to contextualize ratings and support mixed-methods interpretation.

Analytic Strategy and Model Selection Rationale

The AI Workforce Coverage and Maturity framework advances both linear causal claims and configurational claims about organizational AI maturity. Specifically, the framework asserts that (a) higher maturity in specific domains predicts outcomes, (b) balance across domains functions as a higher-order mechanism, and (c) certain domains act as necessary conditions for others. No single analytic technique is sufficient to test all of these claims. Accordingly, the study adopts a complementary, multi-method analytic strategy, with each method selected based on theoretical alignment rather than methodological convenience.

Structural Equation Modeling as the Primary Analytic Technique

Structural Equation Modeling is recommended as the primary analytic approach because the framework is explicitly latent, multi-dimensional, and theory-driven. Each workforce coverage domain represents a latent construct measured by multiple reflective indicators, and relationships among domains are directional and theoretically specified. SEM allows simultaneous estimation of the measurement model and structural relationships, which is essential for establishing construct validity while testing hypotheses.

SEM is particularly well-suited for testing hypotheses related to overall governance effectiveness, moderation effects, and explanatory power. For example, hypotheses asserting that balanced maturity predicts governance effectiveness require modeling both direct effects and higher-order constructs, which SEM can accommodate. Additionally, moderation hypotheses, such as the claim that governance maturity dampens the risk associated with applied AI workflows, can be tested through interaction terms within the SEM framework.

Importantly, SEM enables explicit modeling of measurement error, which is critical given the complexity of organizational AI maturity constructs. This capability strengthens causal inference and addresses reviewer concerns about construct contamination and inflated relationships. Model fit indices further provide transparency regarding how well the theoretical model aligns with observed data, which is a core expectation in ACM and IEEE empirical research.

Fuzzy-Set Qualitative Comparative Analysis for Necessary Conditions and Imbalance

While SEM is appropriate for testing average effects and linear relationships, it is not well-suited for testing necessary-condition logic or equifinality. Several core claims of the framework assert that certain conditions must be present for others to occur, rather than simply increasing their likelihood. For example, the proposition that baseline workforce enablement is required for applied AI maturity reflects necessary-condition logic, not sufficiency.

Fuzzy-set Qualitative Comparative Analysis is therefore recommended as a complementary analytic technique. fsQCA allows examination of whether high levels of one condition are a subset of another, and whether multiple configurations of conditions lead to the same outcome. This aligns directly with the framework's emphasis on balance, imbalance signatures, and alternative pathways to AI maturity.

fsQCA is particularly appropriate for identifying imbalance configurations, such as organizations with high applied AI maturity and low governance maturity and assessing whether these configurations are systematically associated with negative outcomes. By capturing configurational causality, fsQCA complements SEM and provides stronger theoretical coverage than either method alone. The use of fsQCA also signals methodological sophistication and theoretical intentionality to reviewers.

Multilevel Modeling for Role-Based and Organizational Effects

Multilevel modeling (MLM) is recommended when data are collected from multiple respondents within the same organization. In such cases, responses are nested within organizations, violating the independence assumption of single-level models. MLM allows separation of within-organization perceptual variance from between-organization maturity effects, strengthening inference regarding organizational-level constructs.

MLM is particularly relevant for hypotheses related to role-based perceptual divergence. By modeling respondent role at Level 1 and organizational maturity at Level 2, MLM enables examination of how perceptions systematically differ by role while still estimating organizational effects. Even when sample sizes limit full multilevel modeling, intraclass correlation coefficients can be used to justify aggregation and assess consensus.

Data Analysis Plan

The data analysis follows a structured, multi-stage sequence designed to align analytic techniques with the theoretical claims of the AI Workforce Coverage and Maturity framework. Each stage addresses a distinct inferential objective, progressing from construct validation to hypothesis testing, configurational analysis, and role-based interpretation. This staged approach ensures that measurement validity is established prior to structural inference and that complementary analytic methods are used where theoretical claims extend beyond linear or symmetric relationships.

Stage 1: Measurement Model Validation

The first stage establishes the validity and reliability of the three-domain measurement model corresponding to AI for All, AI for Some, and AI for Few. Confirmatory factor analysis is conducted to assess whether survey items load appropriately onto their theorized latent constructs. Each item is specified as a reflective indicator of its respective domain, consistent with the conceptualization of AI maturity as an underlying organizational capability manifested through observable practices. The measurement model is specified as: $x_i = \lambda_i \eta + \varepsilon_i$ where x_i represents the observed item score, λ_i the factor loading, η the latent domain construct, and ε_i measurement error. Model fit is evaluated using multiple indices, including the comparative fit index, Tucker–Lewis index, root mean square error of approximation, and standardized root mean square residual. Internal consistency is assessed using composite reliability and omega coefficients. Measurement invariance across major role categories is tested sequentially at the configural, metric, and scalar levels prior to any group-based comparisons. Establishing invariance ensures that differences observed across roles reflect substantive divergence rather than measurement artifacts.

Stage 2: Structural Model Testing

Once measurement validity is established, the second stage evaluates the core structural relationships posited by the framework using structural equation modeling. Organizational AI governance effectiveness is modeled as a function of the three workforce coverage domains and their degree of alignment. The baseline structural equation is specified as: $AI_GovEff = \beta_1(All) + \beta_2(Some) + \beta_3(Few) + \beta_4(Balance) + \zeta$ where Balance captures coordinated maturity across the three domains. Balance is operationalized primarily as a second-order latent construct and, in alternative specifications, as the inverse variance across domain scores. This stage tests the central claim that workforce-segmented maturity and alignment explain governance effectiveness more robustly than isolated domain strength.

Stage 3: Moderation and Risk Containment

The third stage evaluates the role of governance maturity as a moderating mechanism that constrains AI-related risk arising from applied AI workflows. Applied AI maturity is expected to increase risk exposure in the absence of sufficient governance capacity, while strong governance maturity is theorized to dampen this effect. This relationship is tested using a latent interaction model specified as: $AI_Risk = \beta_1(Some) + \beta_2(Few) + \beta_3(Some \times Few) + \zeta$. A statistically significant negative interaction coefficient provides support for the claim that governance maturity moderates and reduces the marginal risk associated with increased applied AI use. Latent interaction modeling is used to account for measurement error and preserve interpretability.

Stage 4: Necessary-Condition Testing

Several propositions in the framework assert necessary, rather than sufficient, relationships. In particular, baseline workforce enablement is theorized as a prerequisite for applied AI maturity. To test this asymmetric claim, fuzzy-set qualitative comparative analysis is employed as a complementary method. The necessary-condition relationship is expressed as: $Some \leq All$. Consistency and coverage metrics are reported to assess whether high applied AI maturity occurs only in the presence of high baseline enablement. This stage complements SEM results by evaluating constraint-based logic that cannot be fully captured through variance-based techniques.

Stage 5: Imbalance Configuration Analysis

The fifth stage examines imbalance signatures that arise when applied AI capability outpaces governance capacity. Rather than focusing on average effects, this analysis evaluates whether outcome variability increases as a function of misalignment between domains. This relationship is examined using the functional form: $Var(AI_Outcome) = f(|Some - Few|)$. Higher absolute differences between applied AI maturity and governance maturity are expected to be associated with greater instability, inconsistency, or failure in AI outcomes. This stage operationalizes the framework's claim that imbalance is a structural vulnerability rather than merely a low-maturity condition.

Stage 6: Role-Based Analysis

The final stage investigates perceptual divergence across organizational roles, treating differences in reported AI maturity as theoretically meaningful indicators of alignment or misalignment. Multigroup SEM and, where appropriate, multilevel modeling are used to compare latent means and path coefficients across role categories, including executive leadership, IT and security professionals, applied users, and governance specialists. Measurement invariance is established prior to comparison to ensure interpretive validity. Divergence across roles is interpreted as evidence of uneven governance communication, differential exposure to AI practices, or structural separation between policy and practice.

Table 16. Summary of Data Analysis Stages, Methods, and Hypothesis Alignment

Stage	Analytic Objective	Primary Method	Key Constructs / Expressions	Hypotheses Addressed
1	Validate measurement structure	CFA	$x_i = \lambda_i\eta + \varepsilon_i$	All
2	Test core structural relationships	SEM	$AI_GovEff = \beta_1(All) + \beta_2(Some) + \beta_3(Few) + \beta_4(Balance) + \zeta$	H1a, H1b, H5
3	Evaluate governance as risk moderator	Moderated SEM	$AI_Risk = \beta_1(Some) + \beta_2(Few) + \beta_3(Some \times Few) + \zeta$	H3a, H3b
4	Test necessary-condition logic	fsQCA	$Some \leq All$	H2b
5	Identify imbalance signatures	SEM / Difference analysis	$Var(AI_Outcome) = f(Some - Few)$	H4
6	Assess role-based divergence	Multigroup SEM / MLM	Role $\times$ Domain maturity	H6

Theoretical Contributions and Conclusion

This study advances organizational AI research by reconceptualizing AI maturity as a workforce-coverage equilibrium rather than a linear progression of technical capability. By explicitly differentiating baseline enablement, applied workflows, and specialist governance, the framework explains why organizations with similar AI investments experience divergent outcomes. This contribution moves beyond adoption-centric models and positions AI maturity as a socio-technical capability distributed unevenly across the workforce.

A central theoretical contribution is the introduction of balance as a higher-order maturity construct. Rather than assuming that greater maturity in any single domain produces better outcomes, the framework demonstrates that imbalance is a structural source of risk, instability, and stalled scaling. This reframing aligns AI maturity research with

organizational design and systems theory, emphasizing interdependence rather than accumulation.

The study also advances responsible AI scholarship by operationalizing governance not as policy presence but as organizational capacity that varies by workforce segment and risk exposure. This enables empirical examination of governance effectiveness and connects ethical AI principles to observable practice. The AI Workforce Coverage and Maturity Survey further contributes a rigorously structured measurement instrument that supports diagnostic analysis, role-based segmentation, and cumulative research.

By integrating SEM, fsQCA, and multilevel modeling, the empirical design aligns analytic techniques with theoretical claims, strengthening both internal validity and explanatory depth. Together, these contributions provide a robust foundation for future empirical research on AI governance, workforce capability development, and responsible AI scaling across sectors.

This study offers a comprehensive socio-technical model of organizational AI maturity that captures the realities of decentralized adoption, uneven capability distribution, and governance complexity. It provides scholars with a theoretically grounded, empirically testable framework and offers practitioners actionable insight into where AI maturity efforts succeed, where they fail, and why. Taken together, this study advances a general theory of distributed capability maturity in digitally transformed organizations, reframing maturity not as a linear accumulation of technical assets but as an equilibrium achieved through coordinated human, governance, and infrastructural capacity across differentiated roles. While the framework is empirically grounded in artificial intelligence adoption, its underlying logic extends to other AI-adjacent and analytically intensive technologies, including enterprise analytics platforms, algorithmic decision-support systems, and large-scale data infrastructures, where opacity, probabilistic behavior, and uneven workforce engagement similarly challenge traditional maturity models. By explicitly linking capability development to workforce segmentation, risk escalation, and governance proportionality, the framework contributes to organizational design and control literatures by demonstrating how responsibility, accountability, and oversight must be structurally embedded rather than retrofitted. This perspective positions distributed capability maturity as a core organizational design problem in contemporary digital transformation, offering a unifying theoretical lens that connects information systems, governance, and organizational theory in the study of complex socio-technical systems.

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Appendix A. Robustness Checks and Integration of Fuzzy-Set Qualitative Comparative Analysis

A.1 Purpose of Robustness and Complementary Analyses

The primary analyses in this study rely on structural equation modeling to test theory-driven hypotheses concerning workforce coverage, balance, and governance capacity. While SEM is well suited for estimating linear, additive, and moderated relationships among latent constructs, several core theoretical claims advanced by the AI Workforce Coverage and Maturity framework extend beyond correlational logic. In particular, the framework posits necessary conditions, equifinality, and imbalance signatures that are not fully captured by variance-based modeling alone. Accordingly, this appendix documents a series of robustness checks and complementary analyses designed to (a) assess the stability of SEM findings under alternative specifications, (b) examine the sensitivity of results to measurement and aggregation choices, and (c) test necessary-condition and configurational claims using fuzzy-set qualitative comparative analysis. Together, these analyses strengthen internal validity and demonstrate methodological alignment with the underlying theory.

A.2 SEM Robustness Checks

A.2.1 Alternative Measurement Specifications

To assess the robustness of the latent constructs, alternative measurement models were estimated. First, domain-level constructs were re-estimated using reduced indicator sets that excluded outcome-oriented items to ensure that predictive relationships were not mechanically driven by overlapping measurement content. Second, subdomain-level latent constructs were modeled separately to verify that observed relationships were not dependent on specific item groupings. Across specifications, factor loadings remained stable and statistically significant, and substantive conclusions regarding hypothesis support were unchanged. These results indicate that findings are not sensitive to minor variations in measurement structure.

A.2.2 Alternative Balance Operationalizations

Balance across workforce coverage domains was operationalized using two complementary approaches. In the primary analysis, balance was modeled as a second-order latent construct capturing coordinated maturity across AI for All, AI for Some, and AI for Few. As a robustness check, imbalance was alternatively operationalized as the absolute difference between applied AI maturity and governance maturity. Results from both approaches were directionally consistent, with higher imbalance associated with reduced governance effectiveness and increased outcome variability. This convergence supports the theoretical claim that alignment, rather than absolute maturity in any single domain, is a key driver of responsible AI outcomes.

A.2.3 Aggregation and Role Sensitivity

Given the multi-respondent design, robustness checks were conducted to assess sensitivity to aggregation rules. Models were re-estimated using alternative aggregation strategies, including simple means, role-weighted means, and consensus-based inclusion thresholds. Substantive conclusions were consistent across aggregation approaches, indicating that results are not driven by any single role group or aggregation assumption.

A.3 Rationale for fsQCA Integration

Several propositions advanced in this study are explicitly asymmetric and configurational. For example, the claim that baseline workforce enablement is a necessary condition for applied AI maturity asserts that high applied maturity does not occur in the absence of baseline enablement, rather than that baseline enablement increases applied maturity on average. Such claims are not adequately tested using SEM alone. Fuzzy-set qualitative comparative analysis was therefore employed as a complementary method to examine necessary-condition logic and to identify multiple sufficient configurations leading to high or low AI governance outcomes. The integration of fsQCA aligns with prior organizational and information systems research that combines variance-based and set-theoretic approaches to capture complex causality.

A.4 fsQCA Calibration and Conditions

Survey scale scores were calibrated into fuzzy-set membership scores using theoretically informed thresholds. Full membership was assigned to observations with domain scores in the upper maturity band, crossover points corresponded to established maturity, and full non-membership reflected initiating or absent practices. Calibration thresholds were subjected to sensitivity testing to ensure robustness of results. Conditions included baseline workforce enablement, applied AI maturity, governance maturity, and balance indicators. Outcomes examined included high governance effectiveness and low AI risk exposure.

A.5 Necessary-Condition Analysis

Necessary-condition analysis was conducted to test the proposition that baseline workforce enablement is required for applied AI maturity. Consistency scores exceeded accepted thresholds for necessity, while coverage levels indicated substantive relevance. No instances of high applied AI maturity were observed in the absence of high baseline enablement, providing empirical support for the necessary-condition claim. These results complement SEM findings by demonstrating that the enablement relationship is not merely correlational but structurally constraining.

A.6 Configurational Sufficiency and Imbalance Signatures

fsQCA sufficiency analysis identified multiple configurations associated with high governance effectiveness. Configurations characterized by high applied AI maturity and high governance maturity consistently produced positive outcomes, regardless of baseline enablement variation beyond the necessary threshold. Conversely, configurations marked by high applied AI maturity and low governance maturity were associated with elevated risk and unstable outcomes, even when baseline enablement was strong. These imbalance signatures align with the framework's central claim that applied AI capability without commensurate governance capacity represents a structural vulnerability.

A.7 Integration of SEM and fsQCA Findings

SEM and fsQCA findings are mutually reinforcing. SEM establishes average directional effects and moderation relationships, while fsQCA confirms necessity, identifies alternative pathways, and reveals configurations associated with failure. Together, these methods provide a comprehensive empirical evaluation of workforce coverage, balance, and governance alignment. This integrated approach strengthens confidence in the theoretical model by demonstrating that key claims hold across analytic paradigms with different assumptions about causality.

A.8 Summary

The robustness checks and fsQCA analyses reported in this appendix provide strong support for the stability and validity of the primary findings. By triangulating across methods, operationalizations, and aggregation strategies, the study demonstrates that the AI Workforce Coverage and Maturity framework captures durable organizational patterns rather than artifacts of modeling choice. This methodological rigor enhances both the credibility and generalizability of the conclusions.

Appendix B: AI Workforce Coverage and Maturity Survey

Domains: AI for All | AI for Some | AI for Few

Survey Purpose

This survey measures how effectively an organization enables, governs, and improves the use of artificial intelligence (AI) across three workforce domains:

- AI for All: Baseline AI literacy and safe everyday use for the majority of employees.
- AI for Some: Advanced applied use by power users who develop workflows, automate tasks, and apply AI to operational or analytic work.
- AI for Few: Specialist capability required to operate AI at scale, including platform standards, evaluation, governance operations, security, compliance, and lifecycle management.

Survey results are intended to support planning, professional development, risk management, resource allocation, and continuous improvement.

Instructions for Respondents

- Respond based on current practice, not future plans.
- Select the highest level that is consistently true across your area (not based on isolated examples).
- If your area is uneven (some teams mature, others not), rate the typical experience and explain variance in comments.
- Where “Not Applicable” is available, use it only when the capability truly does not apply to your context.

Response Scale (Used Throughout the Survey)

Select one option for each statement:

Option 0 — Not in Place

No established practice exists. Work is absent or entirely informal and inconsistent.

Option 1 — Initiating

Early activity exists (individual use or small pilots), but it is inconsistent, minimally documented, and not broadly understood.

Option 2 — Established

A defined approach exists and is used in multiple settings. Basic documentation, guidance, and repeatable practices are present, but adoption may be uneven.

Option 3 — Operationalized

Standard practices are broadly adopted and supported. Accountability exists, and the capability is integrated into routine operations.

Option 4 — High Performing

Practices are consistently executed at scale, measured with meaningful indicators, and improved through structured feedback and refinement.

Option N/A — Not Applicable (*where provided*)

This item does not apply to my role or to the scope I am rating.

Section 1: Respondent and Context Profile

Section Description

This section establishes the context needed to interpret results, segment findings, and compare maturity across roles and organizational areas.

1.1 Organizational Area

- Department / Unit / Division: _____

1.2 Role Category (Select one)

- Executive Leader
- Dean / Director / Manager / Supervisor
- Faculty
- Classified Professional / Staff
- IT
- Information Security / Compliance / Privacy
- Institutional Research / Institutional Effectiveness / Analytics
- AI / ML Specialist
- Other: _____

1.3 Survey Perspective (Select one)

- Individual role and workflow
- Team/unit/department perspective
- Institution-wide (enterprise perspective)

1.4 AI Use Frequency (Select one)

- Never
- Less than monthly
- Monthly
- Weekly
- Daily or almost daily

1.5 Workforce Domain Alignment (Select one)

AI for All / AI for Some / AI for Few

Section 2: AI for All (Baseline Workforce Enablement and Safe Everyday Use)

Section Description

This section measures the organization's readiness to support broad AI usage for everyday work. It emphasizes policy clarity, responsible use, quality verification, access to approved tools, baseline training, and measurable value.

Rate each statement using the Response Scale (0–4).

2A. Direction, Expectations, and Culture (5 items)

- A1. Employees can describe why the organization supports baseline AI use and how it connects to organizational goals.
- A2. Expectations for AI use are clear and consistent across employee groups (e.g., faculty, staff, managers).
- A3. Employees understand the difference between AI as “assistance” versus AI as a substitute for professional judgment.
- A4. Employees can identify appropriate baseline use cases for their role (e.g., drafting, summarizing, planning).
- A5. Leaders in my area reinforce responsible AI use through communication, modeling, and expectations.

2B. Responsible Use, Policy Clarity, and Risk Awareness (5 items)

- A6. Employees understand what information must never be entered into AI tools (e.g., student records, HR details, credentials, confidential data).
- A7. Employees know how to handle uncertain outputs (e.g., hallucinations, incomplete answers, conflicting claims).
- A8. There is clear guidance on attribution and transparency (e.g., when AI assistance should be disclosed).
- A9. The organization provides guidance on bias and fairness risks in AI-generated content and recommendations.
- A10. There is a clear and used pathway to report AI-related concerns, incidents, or policy questions.

2C. Quality, Verification, and Work Product Standards (5 items)

- A11. Employees consistently verify AI outputs before relying on them for work decisions or external communications.
- A12. Employees use quality practices for AI-assisted writing (accuracy, completeness, tone, and alignment with policy).
- A13. Employees use quality practices for AI-assisted analysis (checking assumptions, methods, and data interpretation).
- A14. Employees understand and apply guidance for citations or sourcing when AI generates factual claims.
- A15. There is a standard expectation that a human remains accountable for the final work product.

2D. Approved Tools, Access, and User Support (5 items)

- A16. Approved AI tools (or approved configurations) are clearly identified for work use.
- A17. Employees understand which tools are not approved and why (privacy, security, or contractual risks).
- A18. Employees can access support resources (how-to guides, templates, examples) without excessive barriers.
- A19. Help pathways exist for users (e.g., office hours, service desk routing, policy contacts).
- A20. AI access and support are equitable across roles and locations (not limited to a small subset of employees).

2E. Baseline Skills, Training, and Readiness (5 items)

- A21. Baseline training is available and designed around practical job tasks (not only conceptual content).
- A22. Baseline training includes privacy, security, bias, and responsible use expectations.
- A23. Baseline training includes accessibility expectations for AI-assisted materials (when applicable).
- A24. The organization communicates updates when tools, risks, or policies change.
- A25. The organization can assess baseline readiness (e.g., training completion, knowledge checks, competency indicators).

2F. Adoption, Benefits, and Continuous Improvement (5 items)

- A26. The organization has identified a set of recommended baseline use cases by function/role.
- A27. Adoption is encouraged through practical supports (templates, examples, coaching, recognized practices).
- A28. The organization tracks at least one benefit indicator (time saved, improved quality, reduced backlog, service consistency).
- A29. The organization gathers feedback from users to improve guidance, training, and tool support.
- A30. Baseline AI use is periodically reviewed and refined to reduce risk and increase effectiveness.

Optional comment for Section 2: What is the primary barrier to improving “AI for All” maturity?

Section 3: AI for Some (Power Users and Applied AI for Workflows and Analytics)

Section Description

This section assesses maturity for advanced applied AI use, including workflow design, automation, analytic augmentation, and department-level deployment. It emphasizes governance for higher-impact uses, documentation standards, controlled piloting, change management, and performance measurement.

Rate each statement using the Response Scale (0–4).

3A. Use Case Identification and Prioritization (5 items)

- S1. My area has a consistent approach to identifying AI opportunities (service delays, process bottlenecks, quality gaps).
- S2. My area prioritizes AI use cases using agreed criteria (impact, feasibility, risk, alignment to goals).
- S3. My area maintains a visible list of AI use cases (ideas, pilots, in production, retired).
- S4. Prioritization includes equity, accessibility, and unintended impact considerations where relevant.
- S5. My area has clear ownership for each AI use case (business owner and operational accountability).

3B. Governance, Oversight, and Human Review (5 items)

- S6. Higher-impact uses require review before release (student-facing, advising, eligibility, HR, financial, compliance-sensitive work).
- S7. Human review requirements are clear (what must be reviewed, by whom, and at what stage).
- S8. There is guidance for appropriate automation boundaries (what AI may draft vs. what humans must decide).
- S9. My area has defined risk tiers for AI uses (low/moderate/high) with corresponding controls.
- S10. My area documents decision points where AI is used and ensures humans remain accountable.

3C. Data Readiness, Knowledge Sources, and Documentation (5 items)

- S11. Power users can identify authoritative data sources and metric definitions (data dictionary, KPI standards, caveats).
- S12. AI workflows document inputs, prompts/configurations, validation steps, and intended use.
- S13. AI workflow documentation includes limitations, exceptions, and known failure modes.
- S14. Version control exists for AI workflows (what changed, why, when, and by whom).
- S15. My area maintains approved knowledge sources for AI-assisted work (policies, FAQs, playbooks) and updates them.

3D. Tooling, Integration, and Controlled Testing (5 items)

- S16. Power users have approved tools to build AI-enabled workflows (low-code automation, APIs, connectors, approved plugins).
- S17. My area can pilot AI workflows safely (limited release, sandbox environment, controlled dataset).
- S18. Testing includes performance checks relevant to the use case (accuracy, completeness, reliability, usability).
- S19. There is a rollback/contingency plan if an AI workflow causes errors or operational disruption.
- S20. My area can monitor AI workflow behavior during pilots and make rapid adjustments.

3E. Skills, Enablement, and Change Management (5 items)

- S21. There is a defined “power user” pathway (competencies, training sequence, mentoring, and peer community).
- S22. Power users receive guidance on communicating changes to stakeholders (frontline staff, supervisors, faculty, students where relevant).
- S23. Operational changes include job aids, training, and support materials for impacted users.
- S24. My area allocates time and capacity for continuous improvement (not only initial build).
- S25. Ethical and responsible use expectations are reinforced for power-user implementations (privacy, fairness, transparency).

3F. Measurement, Outcomes, and Improvement (5 items)

- S26. AI workflows are measured with meaningful metrics (quality, speed, error rate, service outcomes, satisfaction).
- S27. User feedback loops are captured and reviewed (frontline staff, students, stakeholders).
- S28. Benefits and risks are reviewed together (value outcomes plus compliance and quality indicators).
- S29. My area can demonstrate at least one successful AI workflow improvement with documented outcomes.
- S30. My area has a repeatable process to retire, replace, or redesign AI workflows that do not perform adequately.

Optional comment for Section 3: Which governance or operational control is most needed to scale “AI for Some”?

Section 4: AI for Few (Specialist Operations for Platform, Security, Evaluation, and Governance at Scale)

Section Description

This section assesses specialist capability needed to operate AI responsibly and reliably across the organization. It measures enterprise operating model maturity, tool/system inventory management, evaluation and assurance, security and privacy controls, data and knowledge governance, deployment standards, monitoring, and lifecycle management.

Rate each statement using the Response Scale (0–4).

4A. Enterprise Operating Model and Accountability (5 items)

- F1. Enterprise AI roles, decision rights, and escalation paths are clearly defined and communicated.
- F2. There is a structured governance forum or routine (e.g., review board) for AI approvals and oversight.
- F3. The organization maintains an inventory of AI tools/systems in use (purpose, owner, scope, risk level).
- F4. The organization defines risk tiers for AI systems and applies appropriate control requirements per tier.
- F5. AI accountability is aligned with operational ownership (who maintains, monitors, and supports each AI capability).

4B. Evaluation, Assurance, and Responsible AI Controls (5 items)

- F6. Higher-impact AI uses undergo structured evaluation before deployment (accuracy, failure modes, bias where relevant).
- F7. There are documentation standards for AI systems (intended use, limitations, data considerations, operating constraints).
- F8. The organization has testing approaches to identify vulnerabilities or misuse patterns (including adversarial or stress testing where appropriate).
- F9. There are defined acceptance criteria (“release gates”) that must be met before production deployment.
- F10. The organization reviews AI performance periodically and re-evaluates systems as conditions change.

4C. Security, Privacy, Compliance, and Procurement Controls (5 items)

- F11. AI tools/vendors are reviewed through defined security, privacy, and contractual processes before approval.
- F12. Access control and authentication standards apply to AI tools (role-based access, SSO where applicable, least privilege).
- F13. Logging and auditability are implemented for higher-risk AI uses (monitoring who used what, when, and how).
- F14. Incident response processes explicitly include AI-specific scenarios (data leakage, unsafe outputs, compromised credentials, misuse).
- F15. Compliance requirements are integrated into AI governance (privacy obligations, retention, accessibility, regulatory constraints).

4D. Data Governance and Knowledge Stewardship for AI (5 items)

- F16. Data used to support AI is governed (quality checks, lineage/provenance, access permissions, retention).
- F17. Data definitions and authoritative sources are maintained for enterprise reporting and AI-enabled analytics.
- F18. Knowledge sources used by AI (policies, procedures, FAQs) have assigned stewards and update cycles.
- F19. Controls exist to reduce the use of outdated, unapproved, or low-quality content in institutional AI outputs.
- F20. There is a process to handle corrections when AI outputs are found to be wrong or harmful (content updates, policy updates, retraining guidance).

4E. Platform Standards, Architecture, and Deployment Patterns (5 items)

- F21. The organization defines standard AI architectures/patterns (approved models, vendors, configurations, integration approaches).
- F22. Environments are separated appropriately (development, testing, production) for higher-impact AI capabilities.
- F23. The organization maintains secure integration practices (API management, secrets handling, connector governance).
- F24. AI system changes follow change management standards (review, testing, documentation, release coordination).
- F25. The organization has standards for accessibility and user experience for AI-enabled tools deployed to employees or students.

4F. Lifecycle Management, Monitoring, Resourcing, and Improvement (5 items)

- F26. There are defined criteria for moving AI from prototype to production and into steady-state operations.
- F27. AI systems are monitored post-deployment (usage, performance drift, quality issues, risk events, operational reliability).
- F28. The organization can pause, roll back, or disable AI capabilities quickly when risk or failure is identified.
- F29. Specialist capacity is sufficient across required functions (AI/ML, data engineering, security, privacy, accessibility, legal/compliance).
- F30. The organization uses monitoring and incident learnings to refine policy, training, controls, and platform standards.

Optional comment for Section 4: What specialist capability gap is most limiting enterprise maturity?

Section 5 (Optional): Equity, Accessibility, and Education-Specific Practices (Expanded)

Section Description

This optional section provides deeper insight into AI practices that affect teaching, learning, student services, accessibility compliance, and equitable outcomes. It is especially relevant in education environments.

Rate each statement using the Response Scale (0–4).

- E1. AI-assisted content is reviewed for accessibility (readability, alternative text practices, formatting, captioning where relevant).
- E2. Student-facing AI-assisted communications are reviewed for tone, clarity, inclusiveness, and unintended bias.
- E3. AI-enabled outreach/advising processes are assessed for differential impacts across student populations where appropriate.
- E4. There are clear instructional guidelines for student AI use (acceptable assistance, disclosure expectations, prohibited use).
- E5. Tools used in instruction undergo privacy/terms review and are approved through a defined process.
- E6. Faculty and staff have guidance on designing assessments that remain valid in an AI-enabled environment.
- E7. AI used in analytics/IR follows reproducibility expectations (documented methods, validation, versioning).
- E8. AI-enabled recommendations that affect students are explainable at an appropriate level for the user (what factors were considered, limitations).
- E9. Accessibility requirements are integrated into AI procurement and tool deployment decisions.
- E10. The organization has a human override and correction pathway when AI contributes to a service outcome or recommendation.
- E11. Employees receive guidance on handling sensitive student situations where AI is inappropriate or must be tightly controlled.
- E12. Feedback and complaint mechanisms exist for stakeholders impacted by AI-enabled processes.

Section 6: Open-Ended Items (Qualitative Evidence for Planning)

Section Description

These questions capture context behind the ratings and identify actionable priorities, barriers, and resource needs.

- Q1. Identify up to three AI use cases you believe should be scaled within the next 12 months and briefly explain why.
- Q2. Identify the top barriers or risks limiting effective AI adoption in your area.
- Q3. What would most effectively move your area up one maturity level within 6–12 months?
- Q4. What specific support do you need from leadership or central services?

Scoring and Reporting Framework (For Survey Administrators)

Domain Scores (0–4 scale)

- AI for All Score: Average of A1–A30
- AI for Some Score: Average of S1–S30
- AI for Few Score: Average of F1–F30

Optional Equity/Accessibility Score: Average of E1–E12

Interpretation Bands

- 0.00–0.99: Not in Place
- 1.00–1.99: Initiating
- 2.00–2.99: Established
- 3.00–3.69: Operationalized
- 3.70–4.00: High Performing

Optional Overall Index (if you want a single roll-up)

A common workforce-coverage weighting approach:

- AI for All: 50%
- AI for Some: 30%
- AI for Few: 20%

Overall Index = $(0.50 \times \text{All}) + (0.30 \times \text{Some}) + (0.20 \times \text{Few})$